FROM THE QUADRATIC FORMULA TO RAMANUJAN'S CONSTANT ON ONE PAGE

-163 and the almost-integer

One discriminant explains forty primes in a row and a power of e that misses an integer by less than 10^{-12}.

THE RULE

The discriminant decides the roots

The roots of $ax^2 + bx + c = 0$ are

$$x \;=\; \frac{-b \pm \sqrt{\Delta}}{2a}, \qquad \Delta \;=\; b^2 \;-\; 4ac$$

and they are complex when $\Delta < 0$. Euler's prime-producing polynomial $n^2 + n + 41$ has $\Delta = 1 - 164 = -163$, so its roots $n = \tfrac{1}{2}\bigl(-1 \pm \mathrm{i}\sqrt{163}\bigr)$ sit far from the real line. A prime q can divide a value only if -163 is a square modulo q, which fails for every prime below 41, and so the values are prime for every n from 0 to 39, failing first at $n = 40$, where $40^2 + 40 + 41 = 41^2$.

41	43	47	53	61	71
83	97	113	131	151	173
197	223	251	281	313	347
383	421	461	503	547	593
641	691	743	797	853	911
971	1033	1097	1163	1231	1301
1373	1447	1523	1601	1681	$= 41^2$

$n^2 + n + 41$ for $n = 0$ to 40: forty primes, then the first composite.

ALL NINE HEEGNER NUMBERS

Unique factorisation in $\mathbb{Q}(\sqrt{-d})$

Rabinowitsch proved in 1913 that $n^2 + n + p$ is prime for every n from 0 to $p - 2$ exactly when the integers of $\mathbb{Q}(\sqrt{1 - 4p})$ have unique factorisation, or class number one. Gauss conjectured the list of such fields, Heegner proved it complete in 1952 and Baker and Stark confirmed it in 1966–67: $d = 1, 2, 3, 7, 11, 19, 43, 67, 163$, with $d = 4p - 1$ for the odd ones.

d	3	7	11	19	43	67	163
Euler-type polynomial	$n^2 + n + 1$	$n^2 + n + 2$	$n^2 + n + 3$	$n^2 + n + 5$	$n^2 + n + 11$	$n^2 + n + 17$	$n^2 + n + 41$
$-j\bigl(\frac{1+\sqrt{-d}}{2}\bigr)$	0	15^3	32^3	96^3	960^3	5280^3	640320^3
$\mathrm{e}^{\pi\sqrt{d}}$ − nearest integer	-0.24	-0.068	0.14	-0.22	-2.2×10^{-4}	-1.3×10^{-6}	-7.5×10^{-13}

The remaining two, $d = 1$ and $d = 2$, give the Gaussian integers $\mathbb{Z}[\mathrm{i}]$ and $\mathbb{Z}[\sqrt{-2}]$, with $j(\mathrm{i}) = 12^3$ and $j(\sqrt{-2}) = 20^3$.

RAMANUJAN'S CONSTANT

Why $\mathrm{e}^{\pi\sqrt{163}}$ is nearly a whole number

$$\mathcal{R} \;=\; \mathrm{e}^{\pi\sqrt{163}} \;=\; 262\,537\,412\,640\,768\,743.999\,999\,999\,999\,250\,072\,597\,198\,185\,688\ldots$$

The j-function satisfies $j(\tau) = q^{-1} + 744 + 196884\,q + \ldots$ with $q = \mathrm{e}^{2\pi\mathrm{i}\tau}$, and at $\tau = \tfrac{1}{2}\bigl(1 + \mathrm{i}\sqrt{163}\bigr)$, a root of $n^2 - n + 41$, $q = -1/\mathcal{R}$ and class number one forces $j(\tau)$ to be the integer -640320^3. Rearranging,

$$\mathcal{R} = 640320^3 + 744 - \frac{196884}{\mathcal{R}} + \cdots \approx 640320^3 + 744 - 7.4993 \times 10^{-13},$$

which predicts the shortfall to twelve decimal places; smaller Heegner numbers miss by more because their q is less tiny.

POWERS OF $\mathcal{R}$ AND A NUMEROLOGICAL COINCIDENCE

$\mathcal{R}^7$ and the vacuum energy problem

Since $\log_{10}\mathcal{R} = 17.4192$, each power of $\mathcal{R}$ adds 17.42 decades, and $\mathcal{R}^7 \approx 8.60 \times 10^{121}$, equal to 640320^{21} within two parts in 10^{14}. The observed cosmological constant in Planck units is $\Lambda\ell_{\mathrm{P}}^2 \approx 2.9 \times 10^{-122}$, whose reciprocal 3.5×10^{121} lies within a factor of 2.5 of $\mathcal{R}^7$, and the mismatch between the quantum vacuum estimate and observation is quoted as 10^{120}, 10^{122} or 10^{123} depending on the cutoff and on factors of 8π.

The coincidence is striking but carries little weight as evidence. A fixed target is landed within 0.39 decades by some power of $\mathcal{R}$ with probability about $2 \times 0.39/17.42 \approx 4.5\%$, and the freedom to choose the power, the constant and which of several quoted targets to aim at makes such near misses easy to find; no mechanism links the j-function to the vacuum.

n	$n \log_{10}\mathcal{R}$	$\mathcal{R}^n$
1	17.42	2.63×10^{17}
2	34.84	6.89×10^{34}
3	52.26	1.81×10^{52}
4	69.68	4.75×10^{69}
5	87.10	1.25×10^{87}
6	104.52	3.27×10^{104}
7	121.93	8.60×10^{121}
8	139.35	2.26×10^{139}

Conventions and history. The constant is named for Srinivasa Ramanujan (1887–1920), whose 1914 paper on modular equations and approximations to π studies near-integers of the form $\mathrm{e}^{\pi\sqrt{n}}$; Hermite had observed the 163 case in 1859, and the name was popularised by Martin Gardner's April 1975 hoax claiming $\mathcal{R}$ was an exact integer. The polynomial $n^2 + n + 41$ is due to Euler (1772), and j is Klein's modular invariant normalised so that $j(\mathrm{i}) = 1728$.